

INST 327 - Database Design and Modeling

Section Number: 0201

Project Progress Report

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Team Number: 0201 - G6

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Final Project Report

Introduction

During this project we created a database that manages and analyzes electric vehicle population data for the state of Washington. Our primary focus was to create a repository that allows us to access information relating to electric vehicles and plug-in hybrid electric vehicles. The main things within our database will include the vehicles, their make, models, registration locations, and their eligibility for state incentive programs. By storing information such as make, model, model year, electric range, and geographic distribution, the database will serve as a tool for understanding the current state of electric vehicles within the state.

We will concentrate on vehicle registration data from the years 2019 through 2022, while also excluding personal owner information for privacy. The database will be structured to answer specific questions about electric vehicle trends, such as the popularity of different makes and models over time. While the original dataset was exceptionally large, containing records for all electric vehicles registered in Washington, our project will utilize a cleaner version. This curated sample will be representative of the broader population allowing us to demonstrate the database's capabilities without being overwhelmed by the vast amount of data.

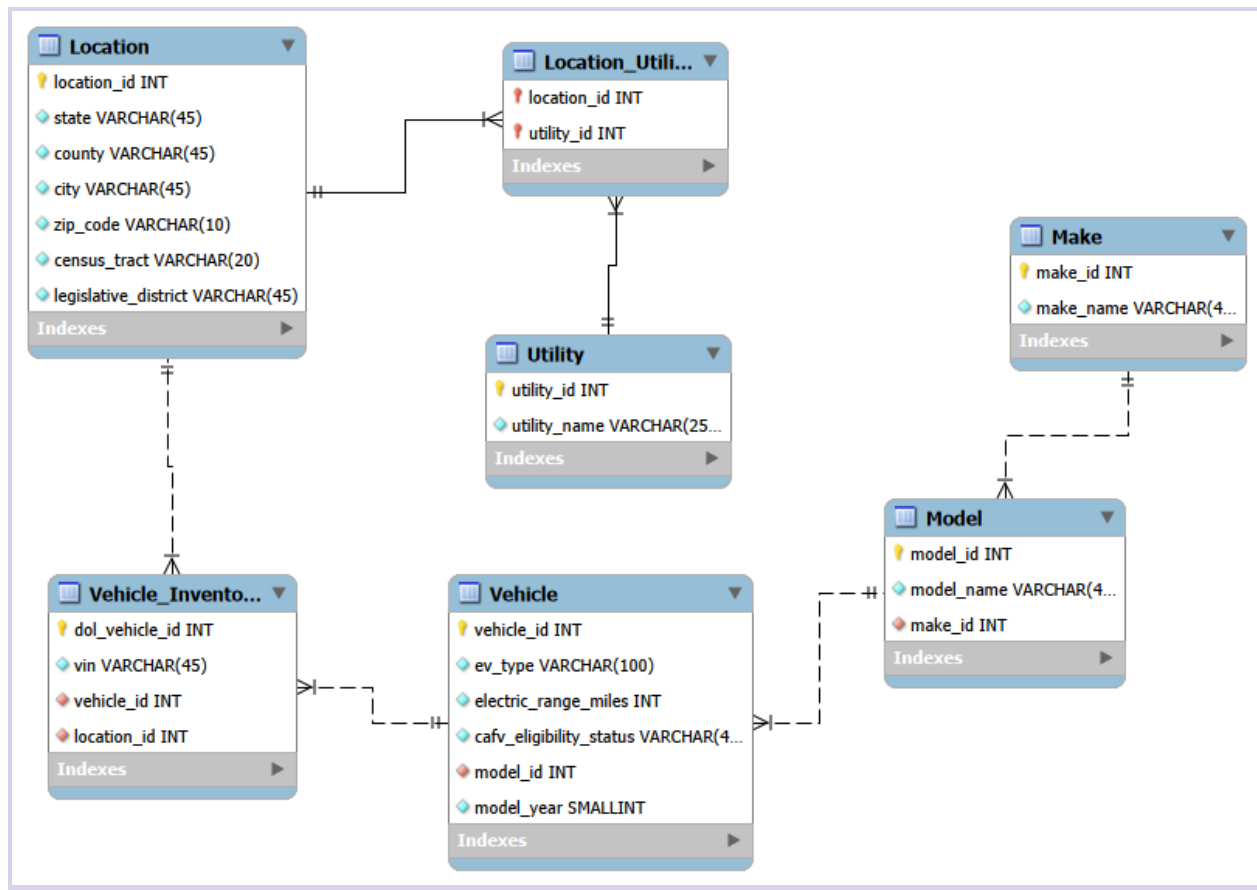
We selected this topic due to the growing popularity of electric vehicles recently and the importance of greener technology in the future. We want to show how data management principles can be applied to support informed decision making for a variety of users

interested in the future

Database Description

Our project will focus on building a database that organizes and analyzes electric vehicle (EV) population data from Washington State using our provided dataset. The dataset consists of information about registered electric vehicles such as vehicle make, model, model year, electric vehicle type (BEV or PHEV), electric range, location (city, county, ZIP code), and CAFV eligibility. The purpose of the database will be to make it easier to analyze EV adoption patterns across different locations over time. Specifically, the database is going to support questions about which vehicles are most common, how EV adoption has changed in recent years, and how EVs are distributed geographically. To keep the project manageable, the scope is limited to vehicles from 2019 through 2023. We are also excluding any unnecessary or sensitive fields from the dataset, especially anything that could be linked to individual vehicle records. This keeps the database focused on population-level trends rather than individual-level data.

Logical Design (include a PNG of your ERD in this section)



Our approach for our database design was to present essential information about electric vehicle make, model, location, and more in a clear and concise way. The tables included information of each different sector of information presented within all of the original database columns. We grouped together all the location related information, individual vehicle information, separated make, model, utility, and inventory. We connected them as needed and made two linking tables. The flow of our information made location and vehicles the two most central and largest tables which make sense as this table in its most basic description is a database of electric vehicles within Washington state. Our tables reach the 3rd normalization form and we didn't find any ways to normalize our information further. We believe this model is a very cohesive design that flows coherently from table to table.

Physical Database

The physical database design had seven normalized tables in total: EV_Type, Make, Model, Location, Utility_Company, Vehicle_Inventory, and Vehicle_Utility. All tables had

primary keys defined. The lookup tables use surrogate autoincremented primary keys whereas Vehicle_Inventory uses the natural key dol_vehicle_id from the original data set. Vehicle_Inventory is referenced by EV_Type, Model, and Location foreign keys.

Once all table schemas were established, we populated the tables with 155 distinct records from the original CSV. Only one duplicate record was removed due to primary key violation. Here are the number of rows in each table: EV_Type (2), Make (19), Model (27), Location (84), Utility_Company (24), and Vehicle_Inventory (155). EV_Type table will have fewer than 15 rows based on its nature since there are only two electric vehicle types. There are five views created and saved in the database in order to showcase JOIN, aggregation, filtering, linking tables, and subquery operations.

Sample Data

The sample data we utilized contained 156 rows, however, one of these rows was a duplicate, so we only utilized 155 unique rows of data. The columns utilized in our data include make, model, model year, electric vehicle type (BEV or PHEV), electric range, MSRP , city, county, and ZIP code, CAFV eligibility. These columns include vehicles from the years 2019 to 2022. There are 27 different vehicle models spanning across 19 different manufacturers. The population of this database is central to Washington state, and our sample from this population includes 65 cities across 21 counties.

Views / Queries

Five SQL views were created to demonstrate the Read part of CRUD using **SELECT** queries. Each view was saved in the database and included in the final database backup. None of the views use the wildcard character *; each query lists the needed columns directly.

The **view_vehicle_inventory_details** view displays detailed information on electric vehicle inventory, including VIN, make, model, model year, EV type, electric range, CAFV eligibility status, and location. This view uses multiple joins and filters vehicles with an electric range of at least 200 miles.

Similarly, the **view_ev_count_by_make_model** view shows how many vehicles are listed for each make and model. This view uses multiple joins, **COUNT()**, and **GROUP BY**.

In addition, the **view_average_range_by_ev_type** view shows the number of vehicles and average electric range for each EV type. This view uses **COUNT()**, **AVG()**, **GROUP BY**, and a **HAVING** filter.

Moving on, the **view_location_utility_details** view shows which utility providers are connected to each location. This view uses the **Location_Utility** linking table along with the **Location** and **Utility** tables.

Finally, the **view_above_average_range_vehicles** view shows vehicles with an electric range above the database average. This view uses a subquery to compare each vehicle's range to the overall average electric range.

Regarding the data source, the original CSV contained 156 rows. One duplicate DOL Vehicle ID was found during import. Since **dol_vehicle_id** is the primary key in **Vehicle_Inventory**, the duplicate row was not inserted. The final **Vehicle_Inventory** table contains 155 unique records.

Views / Queries Requirement Table

Requirement	view_vehicle_inventory_details	view_ev_count_by_make_model	view_average_range_by_ev_type	view_location_utility_details	view_above_average_range_vehicles
Req. A: Multiple tables with JOIN	X	X	X	X	X
Req. B: Filtering with WHERE or	X		X	X	X

HAVING					
Req. C: Aggregation		X	X		
Req. D: Linking table and both source tables				X	
Req. E: Subquery					X

Changes from the original design

During development, several changes were made to improve the database's structure and functionality. One major change was the addition of the **EV_Staging** table during the import process. This temporary staging table was used to load and clean the CSV data before inserting it into the normalized tables. After the import process was completed, the staging table was removed from the final database design because it was only needed for data processing.

Another change involved modifying some column sizes to better fit the imported data. For example, the **utility_name**, **ev_type**, and **census_tract** columns were expanded because some values exceeded the original character limits. This helped prevent truncation errors during import.

The project also required handling duplicate records in the dataset. The original CSV contained 156 rows. One duplicate DOL Vehicle ID was found during import. Since **dol_vehicle_id** is the primary key in the **Vehicle_Inventory** table, the duplicate row was not inserted. The final **Vehicle_Inventory** table contains 155 unique records.

Finally, five SQL views were added to support the project's CRUD and query requirements. These views were not part of the original design but were added later to demonstrate joins, filtering, aggregation, linking tables, and subqueries.

Database Ethics Considerations

The dataset allows us to analyze EV adoption across different geographic areas, but it does not fully capture the social or demographic factors behind those patterns. Because the data focuses on vehicles and locations rather than individuals, it may not reflect the full diversity of experiences related to EV adoption. For example, differences in EV adoption across cities or ZIP codes may be influenced by income levels, access to charging infrastructure, or housing conditions. These factors are not directly included in the dataset, which limits how fully we can interpret the data. To address this, we will be careful not to overgeneralize our findings. We will present the database as a tool for identifying patterns, while acknowledging that it does not explain all underlying causes of those patterns.

Lessons Learned

This project improved our understanding of database normalization, SQL queries, and the relationship between logical and physical database design. One important lesson learned was the value of normalization in reducing redundancy and improving data organization. Separating the data into related tables, such as **Make**, **Model**, **Vehicle**, **Location**, and **Utility**, made the database more efficient and easier to maintain.

Another lesson learned involved importing and cleaning real-world data. The CSV dataset contained duplicate values and inconsistent field lengths, which caused errors during insertion. Using a staging table simplified the cleaning and transformation steps before loading the data into the final normalized tables.

The project also strengthened our understanding of SQL joins, views, filtering, aggregation, and subqueries. Creating the required views demonstrated how multiple tables can

work together to answer meaningful questions about the data. Additionally, exporting and restoring the database backup showed the importance of database backups and data recovery in real database systems.

Potential Future Work

So far the job we have done has mainly focused on the type of vehicle, and how much range it has. Given more time for future work, we could utilize more of the dataset to look at other important details. For example we could look at vehicle prices and where they are located using different cities, zip codes, or legislative districts. Doing this we could see any potential inequalities of income level across different regions of the state.

We could also make more views that could help consumers to choose which kind of vehicle has the best price to range ratio while also being included in the Clean Alternative Fuel Vehicles government program. As we add more views we could also find ways to better optimize the queries in order to improve the efficiency of the dataset.

References

Washington State Department of Licensing. (2024). *Electric Vehicle Population Data*.

Washington State Open Data Portal. <https://catalog.data.gov/dataset/>